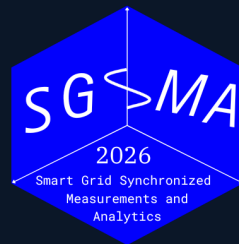


5TH INTERNATIONAL CONFERENCE

Smart Grid Synchronized Measurements & *Analytics*



Physics-Informed PMU Event Detection and Localization

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Date

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Venue

Santiago, Chile



Method overview

Pipeline

1. Convert raw PMU waveforms into physics-aware features.
2. Use a hierarchical ExtraTrees classifier to identify event type.
3. Route each event to a specialized BUS, LINE, or PMU localizer.
4. Produce the challenge CSV output at timestamp resolution.

Input and output

Eight PMU streams go in. For each 30-second window the model returns the event type and where it happened, written at timestamp resolution as the challenge CSV.

Why this design

Physics-aware features and tree ensembles run on a laptop CPU, stay readable, and never look at the hidden test labels.

Problem and Data

PMU streams, event labels, and the prediction contract

Task, event labels, and outputs

Input

Eight synchronized PMU CSV files: one per observed bus, about 90 min at 30 fps, with voltage/current phasors, frequency, ROCOF, DATA_PRESENT, and hidden or blank Event.

Output

TIMESTAMP, Bus, Predicted_Event, and Predicted_Location.

Label	Name	Meaning
0	normal	no disturbance or major data issue
1	fault	fast, strong voltage disturbance
2	line outage	current redistribution without fault-level voltage collapse
3	generation	frequency/ROCOF and power-proxy shift
4	load	sustained current and power change
5	missing	data loss without physical evidence
6	missing + physical	missing data plus physical event
7	bad data	isolated inconsistent sensor/channel
8	unknown	mixed physical evidence not explained by 1–4

Decision structure: detection, event classification, and localization are coupled, not one flat label.

PMU signals and phasor inputs

Raw columns

Group	Columns
Voltage magnitude	VA_MAG, VB_MAG, VC_MAG
Current magnitude	IA_MAG, IB_MAG, IC_MAG
Voltage angle	VA_ANG, VB_ANG, VC_ANG
Current angle	IA_ANG, IB_ANG, IC_ANG
Dynamics	Freq, ROCOF
Quality/time	DATA_PRESENT, TIMESTAMP

Observed PMUs

Buses 2, 5, 6, 10, 19, 22, 29, and 39.

Measurement model

$$V_{\phi}(t) = |V_{\phi}(t)|e^{j\theta_{\phi}^V(t)}, \quad I_{\phi}(t) = |I_{\phi}(t)|e^{j\theta_{\phi}^I(t)}$$

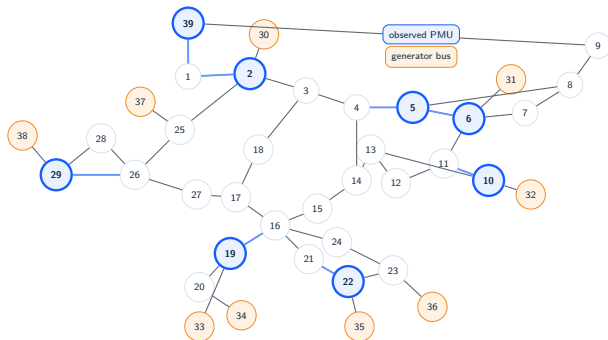
$$\phi \in \{A, B, C\}, \quad \theta_{\text{rad}} = \pi \theta_{\text{deg}} / 180$$

$$\dot{f}(t) = \text{ROCOF}(t)$$

Inference scope

The extractor uses magnitudes, unwrapped angles, frequency, ROCOF, and availability. It does not solve a full AC power-flow at inference time.

IEEE 39-bus system view



Observed PMUs

{2, 5, 6, 10, 19, 22, 29, 39}

Electrical reading

The localizer sees eight sparse PMU views. Events must be inferred from the disturbance pattern across those locations.

Topology used

Electrical distances come from the IEEE 39-bus admittance model, not drawing geometry.

Topology math: from branches to electrical distance

Admittance and impedance

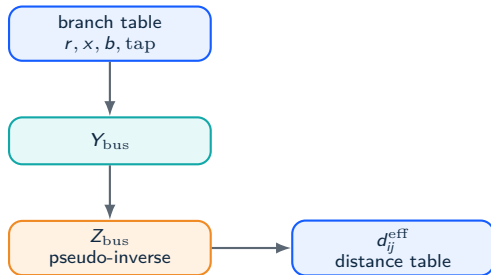
Y_{bus} : network admittance matrix

$$Z_{\text{bus}} = Y_{\text{bus}}^+$$

$$d_{ij}^{\text{eff}} = |Z_{ii} + Z_{jj} - Z_{ij} - Z_{ji}|$$

Meaning

Two buses are close if an injection at one bus is expected to have a similar voltage effect at the other.



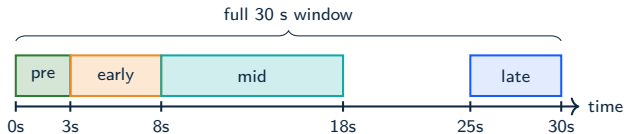
Where it appears

Graph-temporal features use $1/d_{ij}^{\text{eff}}$ weights.

Windowing and Baseline

30-second windows and segment statistics

30-second window and segment design



Segments

$$S_{\text{pre}} = [0, 3), \quad S_{\text{early}} = [3, 8)$$

$$S_{\text{mid}} = [8, 18), \quad S_{\text{late}} = [25, 30)$$

$$S_{\text{full}} = [0, 30)$$

Reason

The window is long enough to capture onset, transient, and settling, but short enough to produce one stable event decision.

Local baseline, robust scale, and subtraction

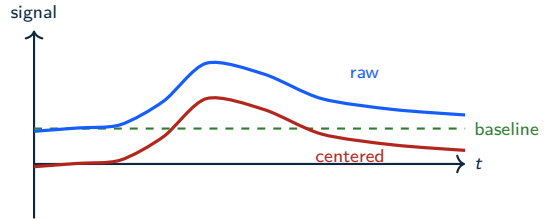
Per PMU and signal

$$b_{p,s} = \text{median}_{t \in S_{\text{pre}}} x_{p,s}(t)$$

$$\sigma_{p,s}^{\text{rob}} = 1.4826 \text{ median}_{t \in S_{\text{pre}}} |x_{p,s}(t) - b_{p,s}| + \epsilon$$

$$x'_{p,s}(t) = x_{p,s}(t) - b_{p,s}$$

$$z_{p,s}(t) = |x'_{p,s}(t)| / \sigma_{p,s}^{\text{rob}}$$



Scale fallback

If MAD is too small, the code falls back to standard deviation and then to a safe value.

Base Feature Engineering

Per-window statistical feature families

Feature schema and base families

Canonical feature key

<PMU/BUS>__<SIGNAL>__<SEGMENT>__<STAT>

Bus2__VA_MAG__early__span

Bus10__Freq__late__mean

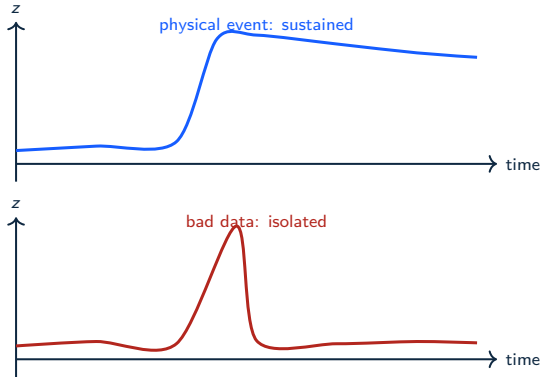
GLOBAL__single_signal_score_max

Meaning

The name stores source, physical signal, temporal segment, and statistic.

Family	Purpose
Segment statistics	magnitude, spread, persistence
Temporal deltas	onset and post-event shift
Derivatives	fast transients and ramps
Robust z-score	scale-normalized abnormality
Sustained anomaly	duration above threshold
Data quality	NaNs, gaps, missing runs
Bad-data score	isolated channel faults
Phase spread	three-phase coherence
Power proxy	rough P/Q movement
Global aggregation	system-wide maxima and counts

Robust z-score separates two failure modes



What it measures

The robust z-score is how far a signal sits from its own pre-event normal, measured in robust units (median and MAD). Scale and noise drop out, so one threshold works across buses and channels.

Why the shape matters

A real event pushes z high and holds it there (top). A bad sample spikes for one frame and falls back (bottom). We give the model both the peak and how long z stays high, so it tells events from sensor glitches.

Data quality, bad data, and phase consistency

Missing-data features

- `n_samples`, `duration_s`, `dt_s`
- `max_timestamp_gap_s`, `max_timestamp_gap_ratio`
- `data_present_fraction`, `data_present_min`
- `missing_run_max_samples/s`
- `nan_fraction_max/mean`

Missing route

Missing only → Event 5. Missing plus physical evidence
→ Event 6.

Bad-data score

$$\frac{\text{top anomaly}}{\text{second anomaly}} \geq 3 \Rightarrow \text{bad data}$$

Phase consistency

Phase-to-phase spread and angle coherence
separate grid events from corrupted single
channels.

Power proxy, global aggregation, and base count

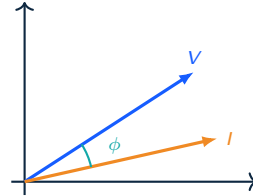
Per-unit magnitudes

$$V_{pu,\phi}(t) = \frac{|V_{\phi}(t)|}{\max(\text{median}_{pre}|V_{\phi}|, \epsilon)}$$

$$I_{pu,\phi}(t) = \frac{|I_{\phi}(t)|}{\max(\text{median}_{pre}|I_{\phi}|, \epsilon)}$$

Proxy, not calibrated power

The challenge data does not require a full power-flow solve. Sign depends on the PMU current convention, but the proxy gives the classifier a stable active/reactive signature.



$$\begin{aligned} P_{\text{proxy}} &= V_{pu} I_{pu} \cos \phi \\ Q_{\text{proxy}} &= V_{pu} I_{pu} \sin \phi \end{aligned}$$

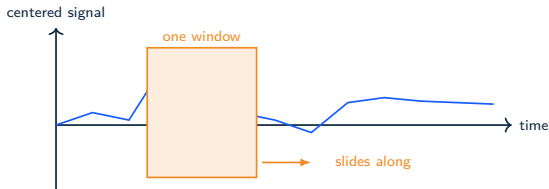
Global summaries and count

Stores cross-PMU mean, max, min, std. Base statistical total: **39,744** features.

Dynamic Feature Engineering

Rolling, adaptive-residual, and graph-temporal evidence

Rolling features: summarize a sliding window



We use window lengths $w = 5, 15, 30, 60, 120$ samples.

Idea

Slide a window over the centered signal and summarize what is inside: mean level, spread, slope, energy, and when the peak arrives.

Why several lengths

Short windows catch fast switching and fault onset. Long windows catch slow load or generation drift. Time-to-peak shows how late the change shows up.

RLS residual: tracking the normal level online

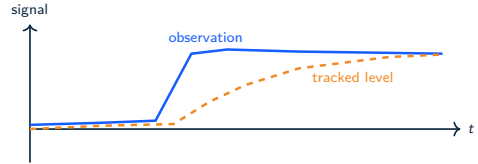
What it does

RLS keeps a running estimate of each signal's normal level and slope and updates it every sample. A forgetting factor λ sets how fast it adapts: we run a fast tracker ($\lambda=0.92$) and a slow one ($\lambda=0.985$).

The residual is the surprise

$$r_t = |y_t - \hat{y}_t|$$

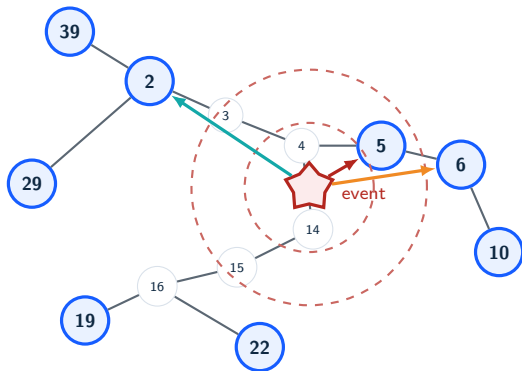
Small while the signal behaves, large the instant it breaks from its recent trend. We keep its peak, mean, p95, and late-minus-pre shift.



Why it helps

A sharp residual spike marks an abrupt event (fault, outage). A slow level change with small residuals marks a ramp (generation, load). Static window statistics miss that timing.

Graph-temporal features: how an event spreads across PMUs



What we read

PMUs sit on different buses. A real event reaches them with a pattern: how much they disagree, which pair differs most, whether they move together, and the timing offset between buses.

Closer buses count more

Each PMU pair is weighted by $1/\text{distance}$, so electrically near buses dominate the score. ExtraTrees turns these into the localization vote.

Hierarchical ML Method

ExtraTrees detection, classification, and event routing

The ML model in one slide

What model

ExtraTrees (Extremely Randomized Trees), from `scikit-learn`.
Each model = `SimpleImputer(median)` + `ExtraTreesClassifier`.
Not a neural network: 0 trainable NN parameters, CPU-only, deterministic.

Four models, same feature row X , different label y

1. Physical event classifier → events 0,1,2,3,4,8
2. Bad-data detector → event 7
3. Missing detector → events 5 vs 6
4. Typed localizers → BUS / LINE / PMU

Why ExtraTrees here

- Handles 45,162 features with no scaling.
- Tolerates NaN via median imputer (real PMU gaps).
- Works with few labeled windows.
- Interpretable: feature thresholds + vote.
- Tiny bundle, fast CPU inference.

Forest vote

$$\hat{y} = \arg \max_c \frac{1}{T} \sum_{t=1}^T p_t(c | X)$$

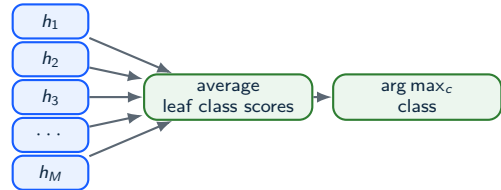
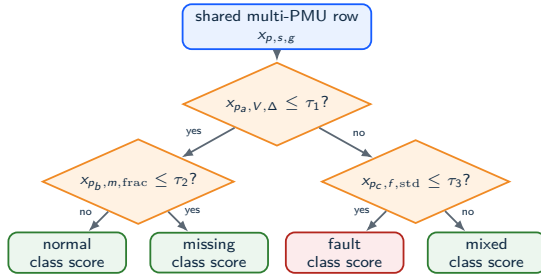
Which feature explains which event

Event	Discriminating feature (physics)	Localizer
1 fault	Voltage span V_{span} + voltage derivative D_V + robust z (fast deep V sag)	BUS (spatial severity)
2 line outage	Current span I_{span} + current-angle change + graph pairwise diff, with V_{guard}	LINE (graph candidates)
3 generation	Freq shift + ROCOF span + P/Q proxy (sustained frequency/power move)	BUS
4 load	Sustained current shift + active-power proxy + rolling energy	BUS
5 missing	data-present fraction <1 , NaN max, timestamp-gap ratio	PMU
6 missing+phys	missing gate <i>AND</i> physical evidence present	physical route
7 bad data	one channel huge robust z , others coherent (isolated spike)	PMU (the sensor)
8 unknown	physical evidence + low confidence for events 1–4	BUS

Four feature families, in plain terms

Window statistics and derivatives measure *abruptness*. Robust z separates *anomaly from noise*. The RLS residual tells a *break from a ramp*. Graph-temporal features show *how the event spreads* across PMUs.

ExtraTrees: one tree and the forest vote



Forest class score

$$s_m(c | x) = \frac{n_{\text{leaf},c}}{n_{\text{leaf}}}, \quad S(c | x) = \frac{1}{M} \sum_m s_m(c | x)$$

$$\hat{c} = \arg \max_c S(c | x)$$

p_a , p_b , and p_c may be different PMUs; each node can also change signal, statistic, and threshold. ExtraTrees averages many randomized trees, so no tree is fixed to one bus or source.

How the ExtraTrees forest is trained

Growing one tree (greedy split, random cuts)

Training rows are labeled windows (x, y) : $x \in \mathbb{R}^{\approx 45k}$ features, y the event class. At every node:

1. draw K random features (not all 45k);
2. pick *one random* threshold for each;
3. keep the cut with the best impurity drop;
4. recurse until leaves are pure or `min_samples_leaf`.

The “Extra” part

Thresholds are *random*, not optimized per feature, and each tree sees the *full* set (no bootstrap). This makes trees fast and diverse, which lowers variance.

Split score: Gini impurity

$$G = 1 - \sum_c p_c^2, \quad \Delta G = G_{\text{node}} - \sum_{\text{child}} \frac{n}{N} G_{\text{child}}$$

Keep the random cut with the largest ΔG .

From tree to forest

Grow M independent trees with different random features and cuts. Each leaf stores class frequencies; the forest averages the leaf probabilities. Tuned controls are `n_estimators`, `max_features`, and `min_samples_leaf`: CPU-only, no scaling, NaN via median imputer.

Why a learned forest, not hand-written if/else

Why if/else on our own features fails

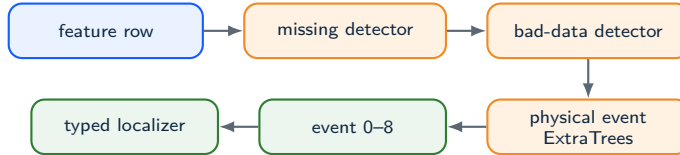
- $\approx 45,000$ features, 9 event classes, 3 localizer types: hand-tuning thresholds is infeasible and brittle.
- One cutoff per feature misses *interactions*: high robust- z and sustained and low missing $\Rightarrow$ fault; high z alone $\Rightarrow$ bad data.
- Hand-picked cutoffs overfit RAW0001 noise and scale, and break on a hidden RAW with different PMUs.

What the forest does instead

- Learns thresholds *and* feature combinations from labeled windows; finds the few discriminative ones among 45k.
- Emits calibrated probabilities $\rightarrow$ feed the physics gates (bad-data, missing) and the confidence checks.
- Tolerates correlated/redundant features and NaN, needs no scaling, and is deterministic, tiny, and fast.

The forest finds threshold combinations across thousands of features that nobody could set by hand, and averages many trees so no single cutoff decides the call.

Hierarchical event routing



Decision order

Missing and bad-data evidence is routed before the physical classifier so communication failures and sensor artifacts do not contaminate physical events.

Event-by-Event Interpretation

How each event type is recognized

Event 1: fault

Main evidence

- Voltage magnitude span.
- Voltage derivative.
- Robust z-score in voltage channels.
- Early segment changes.
- Cross-PMU severity pattern.

Manual recipe

1. Build baseline from pre-event voltage.
2. Detect large voltage sag/transient.
3. Reject missing-only and bad-data-only cases.
4. Localize to BUS using spatial severity.

Event 2: line outage

Main evidence

- Current magnitude span.
- Current derivative.
- Current-angle changes.
- Voltage span not as extreme as a fault.
- Graph-temporal pairwise PMU differences.

Manual recipe

1. Look for current redistribution.
2. Guard against fault-like voltage collapse.
3. Use graph features to score candidate lines.
4. Localize to LINE.

Event 7: bad data

Main evidence

- One signal has extreme robust z-score.
- Top anomaly is much larger than second anomaly.
- Other PMUs/channels remain coherent.
- Missing features do not dominate.

Localizer

Event 7 localizes to PMU, not BUS or LINE. The object that failed is the measurement device/channel source.

Event 8: unknown physical proxy

Main evidence

Event 8 captures cases with physical evidence that does not cleanly fit fault, line outage, generation change, or load change.

Useful features

- Power proxy.
- Rolling dynamics.
- Adaptive residual and innovation features.
- Graph-temporal pattern.

Routing

The final runtime routes Event 8 to a BUS localizer because it represents a physical-grid signature rather than a PMU-only failure.

Localization

Typed localizers and the hybrid bus estimate

Typed origin localization

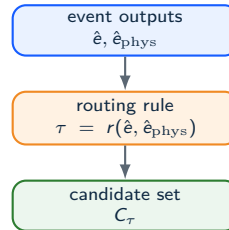
Route before predicting origin

Predicted event	Candidate origin set
0 normal	NONE
1 fault	BUS labels
2 line outage	LINE labels
3 generation change	BUS labels
4 load change	BUS labels
5 missing data	PMU labels
6 missing + physical	physical route
7 bad data	PMU labels
8 mixed physical proxy	BUS labels

For event 6: $\hat{e}_{\text{phys}} = 2 \Rightarrow \text{LINE}$; otherwise BUS.

Implemented decision

The model does not rank every possible label together. It first chooses a type, then runs the localizer trained for that type.



Training, Validation, and Results

Evaluation protocol and reported metrics

Training data and split

Training sources

- 5,000 synthetic SGSMA scenario windows.
- 3,500 train windows and 1,500 stratified holdout windows.
- Base statistical and dynamic tables generated by the repo feature extractors.
- Random state: 20260503.

Validation

- 30% stratified SIM holdout.
- RAW0001 transfer validation.
- Metrics: detector, classifier, macro-F1, localization.

Transfer principle

RAW labels were used for validation and analysis, not to train the final production-feasible submission path.

Final metrics

SIM holdout

Metric	Score
Detector accuracy	0.9693
Classifier accuracy	0.9673
Classifier macro-F1	0.9598
Localizer Top-1	0.8524

RAW0001

Metric	Score
Detector accuracy	1.0000
Classifier accuracy	1.0000
Classifier macro-F1	1.0000
Localizer Top-1	0.6667

Interpretation

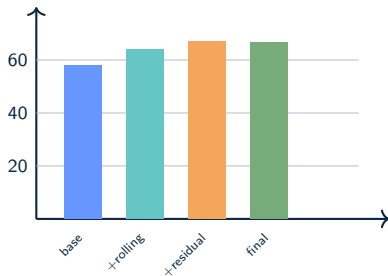
Detection and classification transferred perfectly on RAW0001; localization remained the hard part.

Ablation

What each component contributes, and why we measure it

Ablation evidence

RAW0001 Top-1 (%)



What ablation is

We add one feature family at a time and re-measure localization. Each bar is the score with that group added.

Why it matters

Every group earns its place: rolling, then the RLS residual, then graph features each lift the score, and the full set is best. Nothing is there just for show.

Closing

Submitted method

We turn each PMU window into physics-aware features, classify the event with hierarchical ExtraTrees, and localize it with a typed localizer.

Feature engine

Base statistical features + rolling dynamics + adaptive residuals + graph temporal features.

Classifier

Physical event model + missing detector + bad-data detector + rules.

Localizer

Specialized BUS, LINE, and PMU ExtraTrees localizers.

Selected features: base statistical + rolling + residual/innovation + graph temporal

Conclusion

Core method	Why it worked	Localization
Raw PMU windows are converted into physically meaningful feature rows. Hierarchical ExtraTrees models classify the event type.	Robust statistics capture abnormal behavior. Residual, innovation, and graph-temporal features add timing and network evidence.	Typed localizers keep BUS, LINE, and PMU targets separate. Each model ranks valid candidates inside its own target space.

Reproducible by design

Every output traces back to a feature row, a routing rule, or a tree vote, so any prediction can be explained and rebuilt.